

UNIVERSITAS
GADJAH MADA

Critical Thinking

Metode Penyelesaian Masalah dan Pemodelan

Dr. Indriana Hidayah, S.T., M.T.



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What is Thinking?

Why doesn't SHE like me?

Why doesn't HE like me?



As you start asking questions and seek answers, you are **thinking**.

What is Thinking?

Thinking is one of the primary cognitive skills that distinguishes humans from other creatures.



The ability to think allows us to **understand** the world, make **better decisions**, and develop **innovations**.



Types of Thinking

Studying various types of thinking helps us understand **the best ways to process information**, solve problems, and make decisions in different aspects of life

1. **Critical Thinking**
2. **Creative Thinking**
3. **Analytical Thinking**
4. **Abstract Thinking**
5. **Concrete Thinking**
5. **Divergent / Lateral Thinking**
6. **Convergent / Vertical Thinking**



Types of Thinking

Critical thinking involves asking questions and probing the underlying validity and assumptions behind statements or ideas. Critical thinking is very concerned with investigating the origins of claims and how they come about, and whether they come from hearsay or misinformation.

Creative thinking is always engaged with producing something new: new ideas, new theories, new solutions.

Analytical thinking is about breaking ideas and solutions become component parts and creating methods for categorizing and examining those parts using evidence and logic.

Abstracts thinkers can pull together seemingly disparate ideas together, and they can also reveal the hidden meanings behind concepts.



Types of Thinking

Concrete thinking is also known as practical thinking. Concrete thinking doesn't get carried away with context, analysis, or brainstorming. Its goal is to find the simplest solution as quickly as possible.

Divergent thinking is method is to find as many different solutions as possible in order to spark new ideas. Sometimes you even stray far away from the original problem in order to look for inspiration. This type of thinking is therefore often connected with creative and abstract thinking.

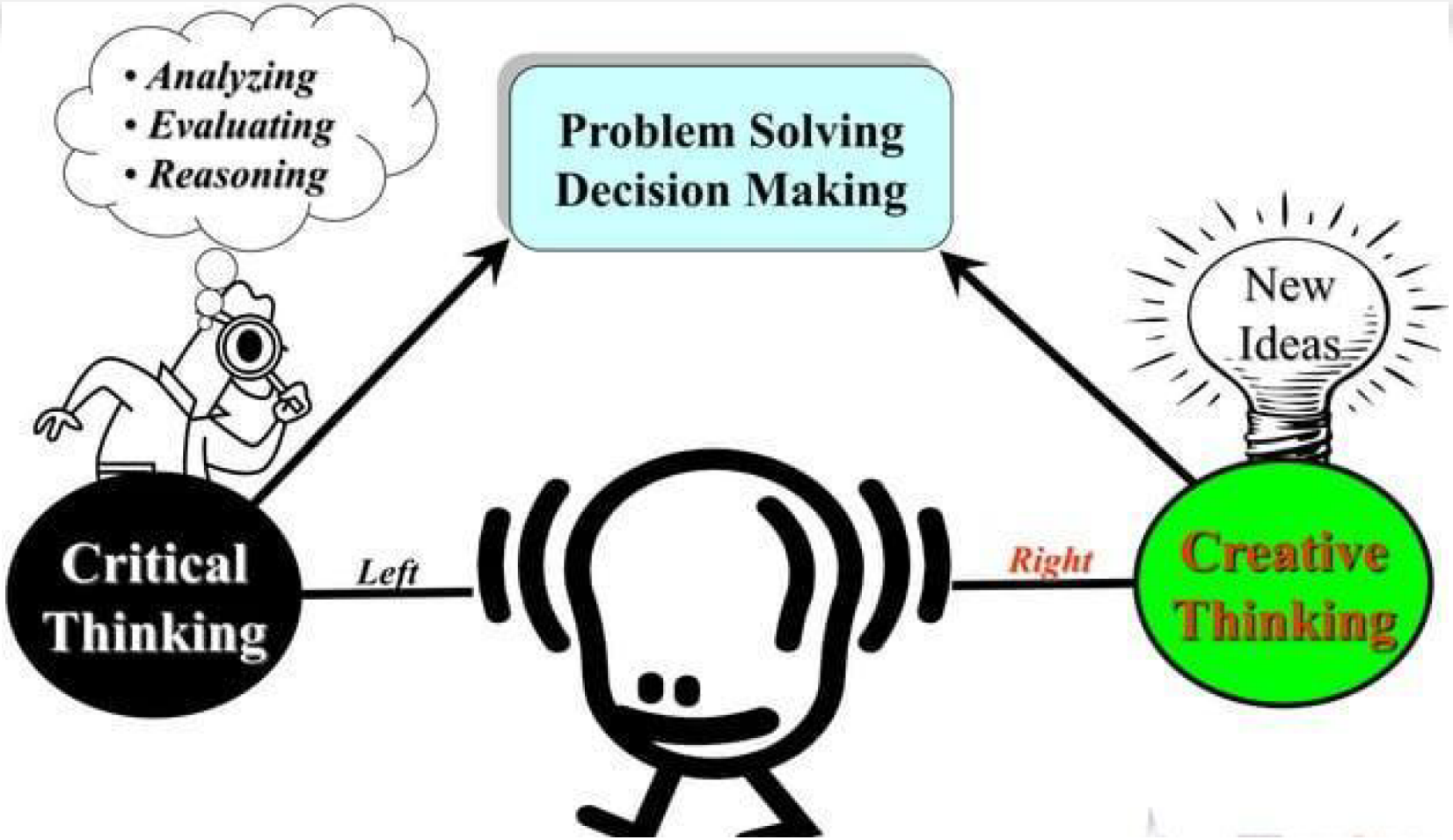
Convergent thinking is about finding one single path forward by combining ideas or eliminating them, through logic and analysis. Convergent thinking is very organized and focused on efficiency, but there is also some elegance and artfulness to it, since it does require combining and merging different ideas together.



Types of Thinking

Studying various types of thinking helps us understand **the best ways to process information**, solve problems, and make decisions in different aspects of life

Types of Thinking



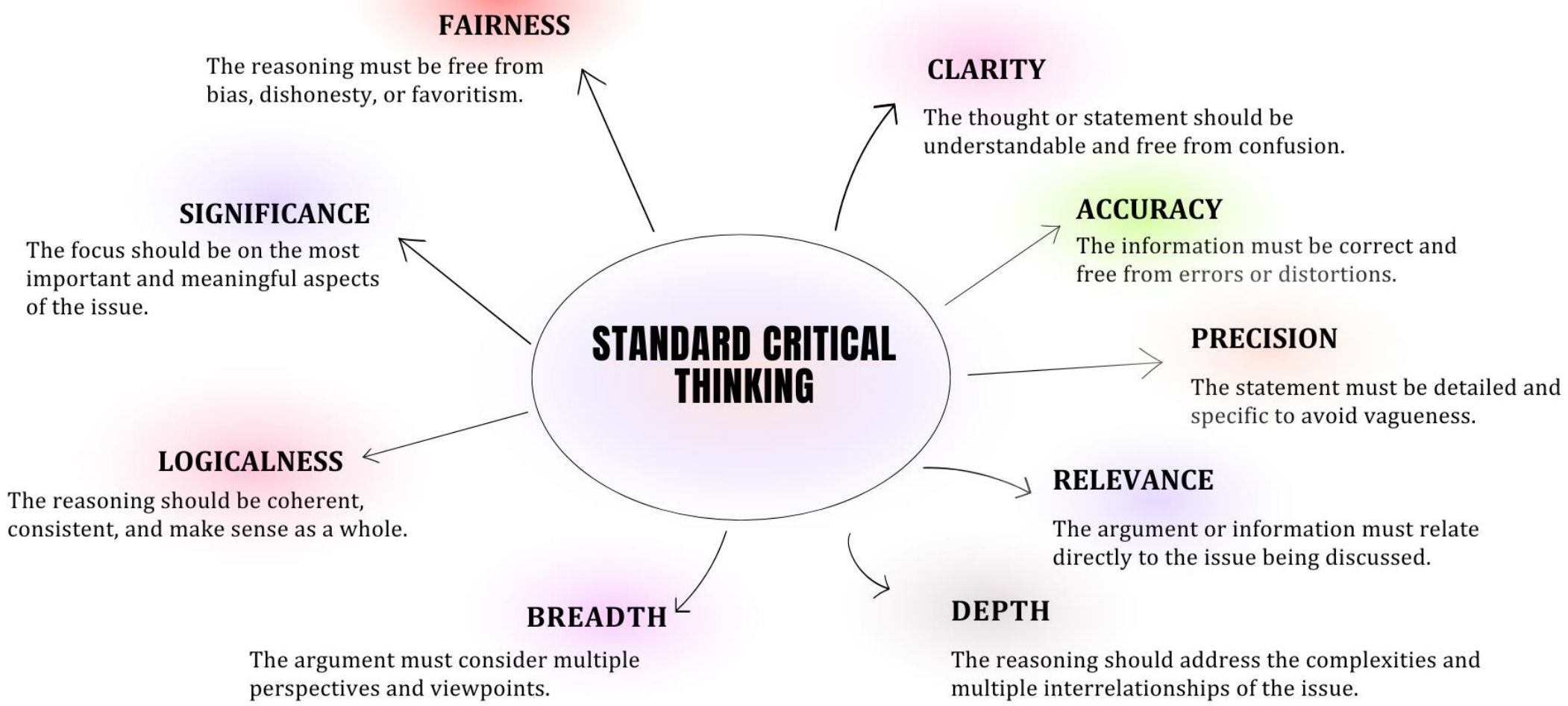


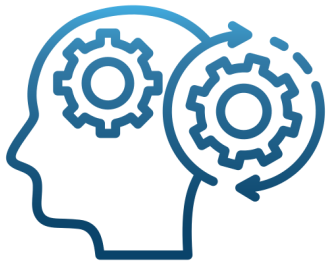
Critical Thinking

Critical Thinking is the general term given to a wide range of cognitive and intellectual skills needed to:

- Effectively identify, analyze, and evaluate arguments.
- Discover and overcome personal prejudices and biases.
- Formulate and present convincing reasons in support of conclusions.
- Make reasonable, intelligent decisions about what to believe and what to do.

Critical Thinking Standards (CTS)





CTS: clarity

- Could you elaborate further on that point?
- Could you express that point in another way?
- Could you give me an illustration?
- Could you give me an example?

CTS: accuracy

- Is that really true?
- How could we check that?
- How could we find out if that is true?



A statement can be clear but not accurate

CTS: precision

- Could you give more details?
- Could you be more specific?



A statement can be both clear and accurate, but not precise

CTS: precision

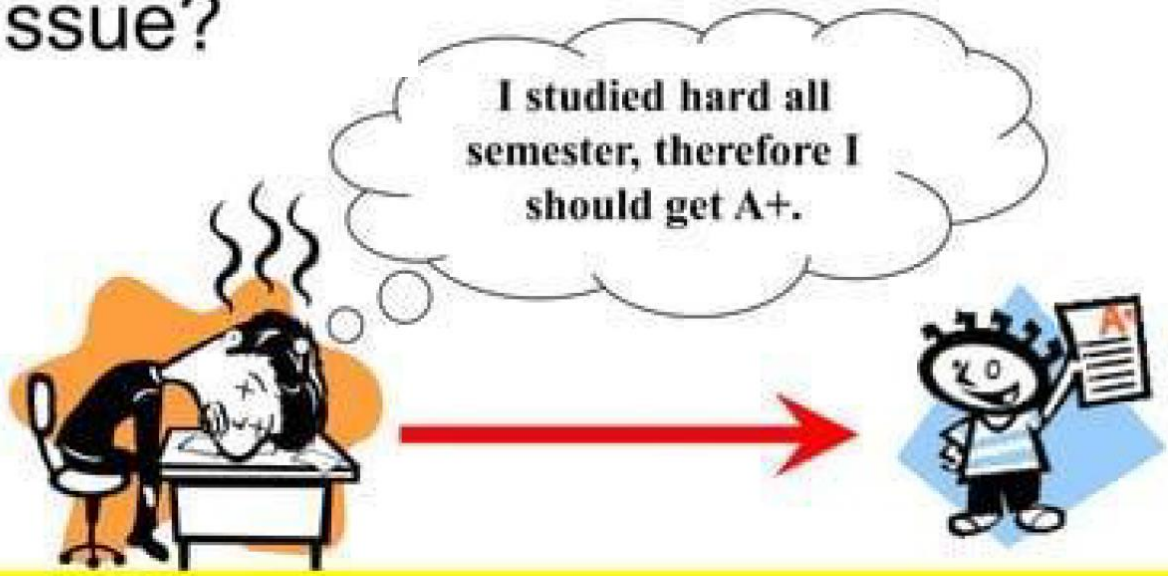
- Could you give more details?
- Could you be more specific?



A statement can be both clear and accurate, but not precise

CTS: relevance

- How is that connected to the question?
- How does that bear on the issue?

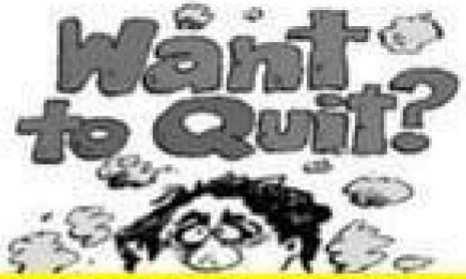


A statement can be clear, accurate, and precise, but not relevant to the question at issue.

CTS: depth

- How does your answer address the complexities in the question?
- How are you taking into account the problems in the question?
- Is that dealing with the most significant factors?

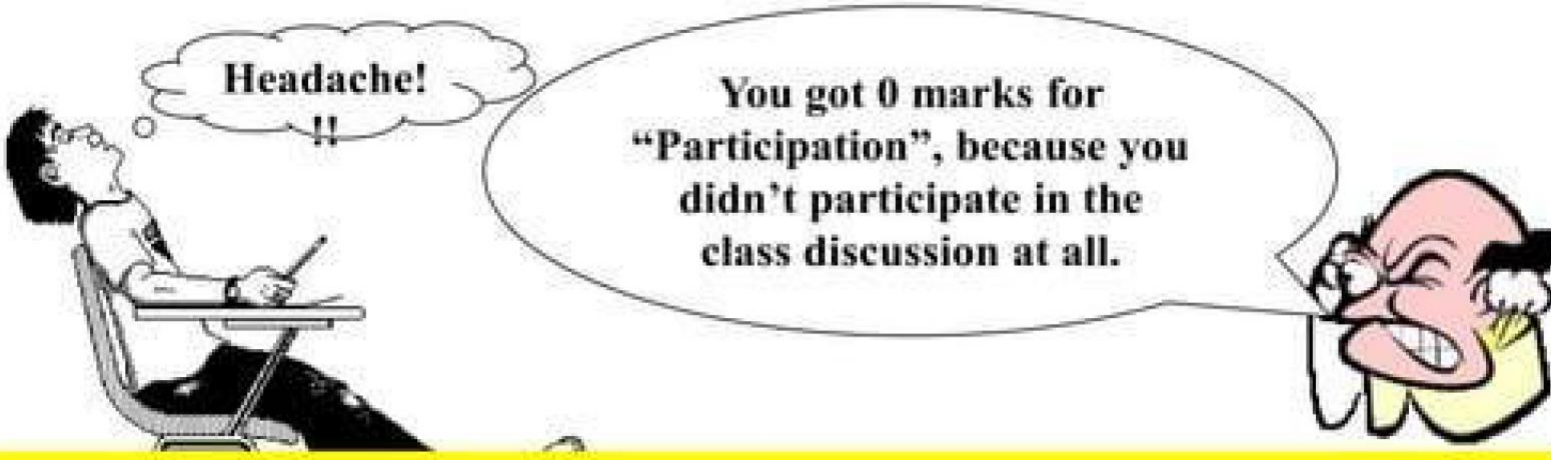




A statement can be clear, accurate, precise, and relevant, but superficial.

CTS: breadth

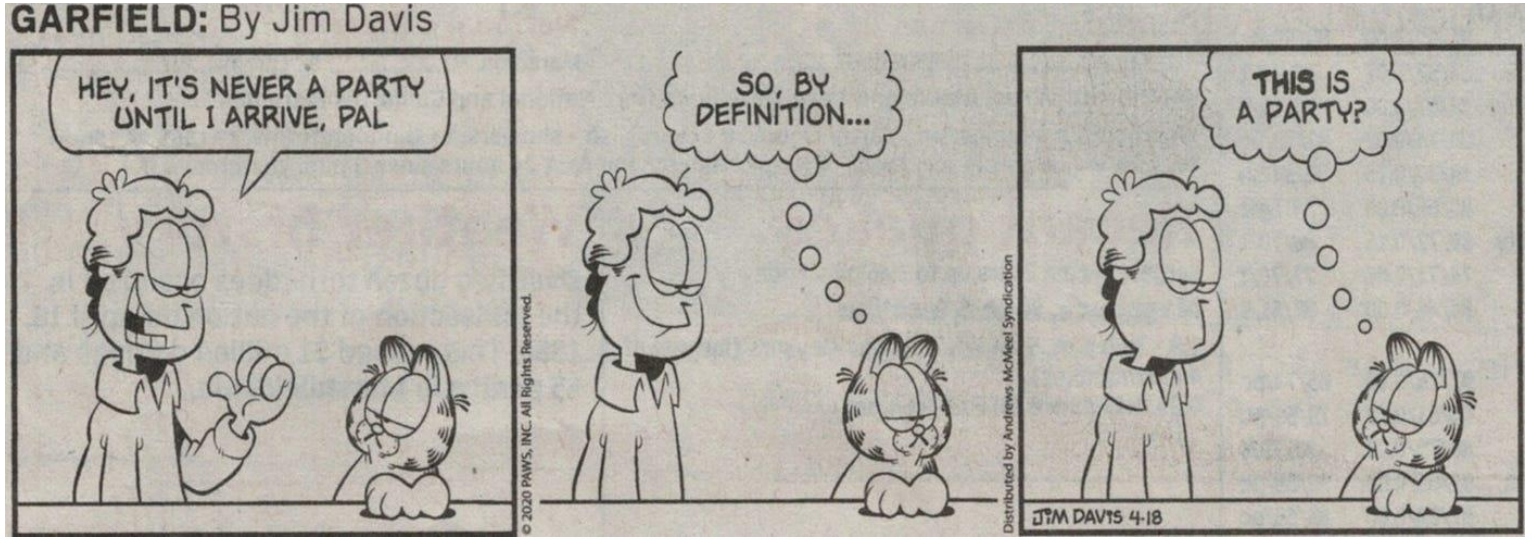
- Do we need to consider another point of view?
- Is there another way to look at this question?
- What would this look like from a conservative standpoint?
- What would this look like from the point of view of...?



A line of reasoning may be clear, accurate, precise, relevant, and deep, but lack breadth.

CTS: logic

- Does this really make sense?
- Does that follow from what you said?
- How does that follow?
- But before you implied this and now you are saying that; how can both be true?



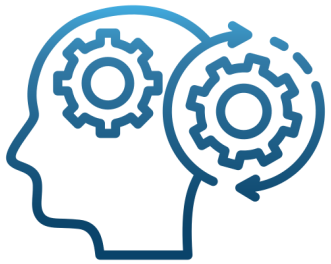
CTS: fair

Critical thinking demands that our thinking be fair.

- Open-minded
- Impartial
- Free of distorting biases and preconceptions



Fair-mindedness is an essential attribute of a Critical Thinker.



Generally, good thinking is…

CLEAR.....rather than.....UNCLEAR

ACCURATE.....rather than.....INACCURATE

PRECISE.....rather than.....VAGUE

RELEVANT.....rather than.....IRELEVANT

CONSISTENT....rather than.....INCONSISTENT

LOGICAL.....rather than.....ILLOGICAL

COMPLETE.....rather than.....INCOMPLETE

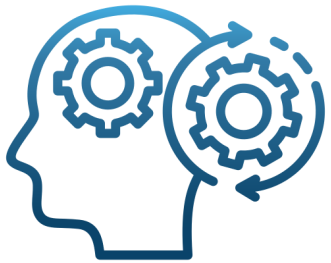
FAIR.....rather than.....BIASED



Benefits of Critical Thinking

Examples:

- **Academic Performance**
  - ❑ understand the arguments and beliefs of others
  - ❑ Critically evaluating those arguments and beliefs
  - ❑ Develop and defend one's own well-supported arguments and beliefs.
- **Workplace**
  - ❑ Helps us to reflect and get a deeper understanding of our own and others' decisions
  - ❑ Encourage open-mindedness to change
  - ❑ Aid us in being more analytical in solving problems
- **Daily life**
  - ❑ Helps us to avoid making foolish personal decisions.
  - ❑ Promotes an informed and concerned citizenry capable of making good decisions on important social, political and economic issues.
  - ❑ Aids in the development of autonomous thinkers capable of examining their assumptions, dogmas, and prejudices.



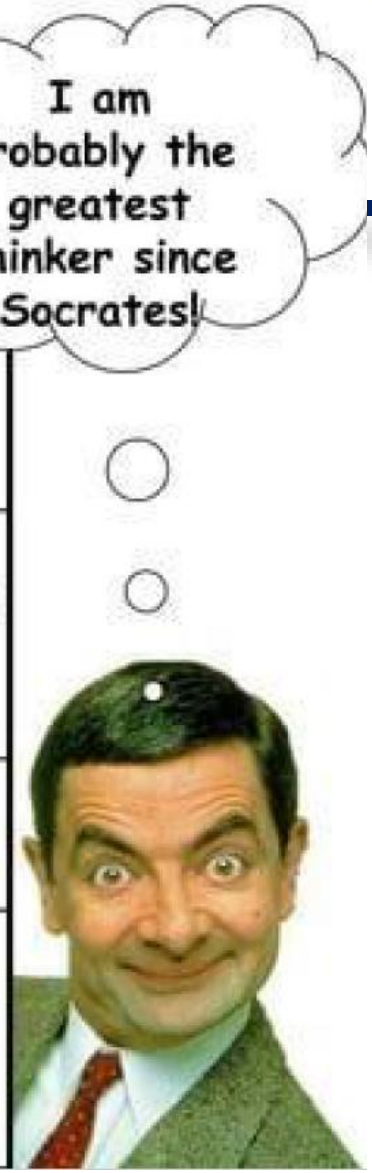
Barriers to Critical Thinking

If Critical Thinking is so important, why is it that ***uncritical thinking is so common***?

Why is that so many people including many highly educated and intelligent people ***find critical thinking so difficult?***

Five *Powerful Barriers* to Critical Thinking:

Egocentrism	**Self-centered thinking** • *self-interested thinking* • *self-serving bias*
Sociocentrism	**Group-centered thinking** • *Group bias* • *Conformism*
Unwarranted Assumptions	**Beliefs that are presumed to be true without adequate evidence or justification** • *Assumption* • *Stereotyping*
Wishful Thinking	**Believing that something is true because one wishes it were true.**
Relativistic Thinking	**The truth is "just a matter of opinion"** • *Relativism* ❑ *Subjectivism* ❑ *Cultural relativism*



Mini quiz

In a 1989 international study of 13-year-olds, Koreans finished first in mathematics and Americans finished last. Yet when asked whether they thought they were "good at mathematics," only 23 percent of Koreans said "yes," compared to 68 percent of Americans.

Which critical thinking barrier do the American students exhibit:

A) Self-interested thinking
B) Group bias
C) Self-serving bias
D) Conformism

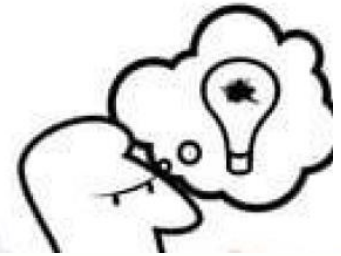


Mini quiz

Muhammad Ali [speaking in Zaire, Africa]: "There's no country as great as the smallest city in America. I mean [here in Zaire] you can't watch television. The water won't even run right. The toilets won't flush. The roads, the cars- there's nothing as great as America."

Which critical thinking barrier does Ali display in this passage?

A) Self-interested thinking
B) Group bias
C) Self-serving bias
D) Conformism



Mini quiz

Adam: My friend Andy is a 1st year student at UNITAR. He is cool, loves hanging out, and has a very laid-back personality.

Lee: I bet he's from KL.

Which critical thinking barrier does Lee exhibit?

A) Self-interested thinking
B) Stereotyping
C) Group bias
D) Conformism



Characteristics of critical thinker

Are you OPEN MINDED about other people's view?

Are you HONEST to yourself (or others) when you are wrong?

Do you have the COURAGE and PASSION to take initiative and confront problems and meet challenges?

Are you AWARE of your own biases and preconceptions?

Do you WELCOME CRITICISM from other peopl

Do you have INDEPENDENT opinions and are not afraid to disa ee?



Critical thinking requires us to use our imagination, seeing things from perspectives other than our own and envisioning the likely consequences of our position

- Bell Hooks

Any questions?